

MediKiosk: Grand Finale Pitch & Defense Manual

Problem Statement ID: 26047 | Ministry of Ayush / AIIA | Team: BINARY BRAINS (GDT007)

■ Presentation Timeline & Time Budget (Target: 5m 45s)

SECTION	TOPIC & SLIDE MAPPING	TIME	RUNNING CLOCK
Section 1	Slide 1: Title & High-Impact Hook	0:20	0:00 - 0:20
Section 2	Slide 2: Problem Framing (The 2-Min OPD Crisis)	0:45	0:20 - 1:05
Section 3	Slide 2 & 3: Solution Walkthrough (5-Step Engine)	1:40	1:05 - 2:45
Section 4	Slide 3: Architecture & Sovereign AI Justification	1:00	2:45 - 3:45
Section 5	Slide 2 & 4: Novelty & AYUSH Differentiation	0:30	3:45 - 4:15
Section 6	Slide 4 & 5: Feasibility, Economics & National Scale	0:45	4:15 - 5:00
Section 7	Live Demo: 45-Second Interactive Walkthrough	0:45	5:00 - 5:45
Section 8	Slide 6: Closing Statement & Call to Action	0:15	5:45 - 6:00

■ Part 1: Word-for-Word Spoken Pitch Script

Slide 1 • [0:00 - 0:20] • Opening Hook & Identity

Respected Jury Members, in clinical medicine, 80% of diagnostic accuracy comes entirely from taking a proper medical history. Yet, according to a landmark study published in BMJ Open, India's average primary-care consultation lasts just over two minutes. I am Potla Sri Sharan, representing Team BINARY BRAINS, and today we present MediKiosk — India's first sovereign AI-powered multimodal clinical intake platform, built specifically to solve the first-mile case-taking bottleneck for the Ministry of Ayush and national healthcare.

Slide 2 • [0:20 - 1:05] • Problem Framing & The 2-Minute OPD Bottleneck

Every single morning, apex government hospitals like AIIMS and AIIA register 4,000 to 10,000 OPD patients. Imagine a doctor with 120 seconds per patient. In that two-minute window, the physician must simultaneously decipher crumpled, handwritten prior paper prescriptions, break language and dialect barriers, perform physical exams, and type clinical notes. In Ayurvedic medicine, the crisis is even more severe. Authentic clinical Ayurveda demands Dashavidha Pariksha — analyzing 10 deep constitution parameters including Prakriti, Agni, Vikriti, and Koshtha. In a crowded OPD, capturing this manually is practically impossible, forcing doctors to abbreviate the very holistic assessment that defines AYUSH. Furthermore, while the Ayushman Bharat Digital Mission (ABDM) has built national digital highways with ABHA IDs, the first-mile data entry problem remains unsolved. Doctors are being turned into data-entry operators.

Slide 2 & 3 • [1:05 - 2:45] • Solution Walkthrough (5-Step Patient Journey)

To solve this, we built MediKiosk — a 5-step, zero-training clinical intake platform that completes the entire case-taking, translation, and document digitization before the patient ever enters the doctor's chamber.

1. Step 1 (Identify & Consent): The patient approaches the kiosk, scans their ABHA QR card or authenticates via Aadhaar OTP, selects their mother tongue, and grants a DPDPA 2023-compliant audio-guided consent.
2. Step 2 (Conversational History & SOCRATES): The patient speaks naturally in their native mother tongue — whether Telugu, Hindi, or Tamil. Our clinical engine listens and dynamically probes onset, character, radiation, and triggers following the clinical SOCRATES ontology. Crucially, if the patient reports crushing chest pain, our Red-Flag Detector instantly triggers Emergency STAT Triage, bypassing routine queuing.
3. Step 3 (Document AI & Lab OCR): The patient places old paper prescriptions or lab reports under the camera scanner. Our Sarvam Vision 3B VLM digitizes cursive handwriting, extracts active medications, and flags abnormal biomarkers — like high Serum Uric Acid — in under 2 seconds.
4. Step 4 (ABDM Linking & Token Generation): The data is structured, bound to the patient's Care Context, and a physical/digital token is issued.
5. Step 5 (Physician Review): The moment Token #061 enters OPD Room 4, the Doctor Portal instantly displays a standardized SOAP note in pure clinical English, complete with the patient's original voice recording, lab highlights, and Ayurvedic Dashavidha radar metrics. The physician reviews in 15 seconds, makes adjustments, and approves it with a single click.

Slide 3 • [2:45 - 3:45] • Technical Architecture & Sovereign AI Justification

Let's address the technical architecture under the hood, and why we engineered it this way: First, a 100% Sovereign Indian AI Stack: We deliberately rejected Western models like GPT-4 or Whisper. They fail on rural Indian accents and violate national health data sovereignty. We integrated Sarvam AI's indigenous stack — Sarvam-105B for clinical reasoning and Dashavidha analysis, Sarvam Bulbul v3 for natural TTS across 11 Indian languages, Sarvam Mayura v1 for medical translation, and Sarvam Vision 3B for cursive prescription OCR. Second, a Zero-Latency Serverless Edge Architecture: The frontend is built with ultra-lightweight Vanilla JS and Tailwind CSS, executing at 60 FPS even on affordable ₹15,000 kiosk terminals. Third, a Dual-Sync Database Engine: Built on Vercel Serverless Edge Functions and Supabase Managed PostgreSQL. We engineered a dual-sync pipeline: state syncs across browser tabs in 0 milliseconds, while encrypted clinical records and NRCeS-compliant FHIR R4 Bundles persist to the cloud with full Row-Level Security.

Slide 2 & 4 • [3:45 - 4:15] • Novelty & AYUSH Differentiation

What makes MediKiosk radically different from existing hospital token machines or tele-triage apps? First, existing hospital kiosks are purely administrative — they only capture a name and print a slip. MediKiosk is a deep clinical history acquisition engine. Second, mobile apps exclude rural, low-literacy, and elderly citizens. MediKiosk requires zero smartphone literacy — every question can be answered by speaking or tapping large touch-chips. Third, it is the only platform built natively for AYUSH. It bridges Allopathic emergency detection with Ayurvedic Dashavidha Pariksha and NAMASTE / WHO ICD-11 Traditional Medicine coding. Finally, Doctor-in-the-Loop: MediKiosk never auto-prescribes or replaces the doctor. It presents a structured draft for physician verification.

Slide 4 & 5 • [4:15 - 5:00] • Feasibility, Economics & National Scalability

Let's talk deployment feasibility and national impact: Hardware Economics: MediKiosk runs on any standard Android touchscreen tablet or commercial POS kiosk terminal equipped with an entry-level document camera and noise-canceling directional mic — costing under ₹25,000 per unit. Clinical Impact: Intake time slashed by 80% — from 15 minutes of manual history-taking down to under 2 minutes; a 3x Increase in Daily OPD Throughput without adding a single extra staff nurse; and 100% ABDM Compliance. Scale-up Roadmap: From AIIA Delhi, MediKiosk can scale through existing e-Hospital and Ayush Grid infrastructure across all 700+ District Hospitals and Ayushman Arogya Mandirs nationwide.

■ Part 2: 45-Second Live Demo Cue Script

TIME	UI ACTION ON SCREEN	SPOKEN NARRATION (SYNCHRONIZED)
0:00 - 0:10	Open Kiosk (/), tap Telugu (■■■■■■■)	"Here is our live deployed Kiosk. The patient selects Telugu. Notice the audio guide speaks in native Telugu using Sarvam Bulbul v3."
0:10 - 0:20	Tap Demo Profile (Ramesh Chandra)	"We authenticate via ABHA ID 91-4829-1029-4821. The DPDPA 2023 audio consent is presented."
0:20 - 0:30	Speak/Tap Knee Pain (■■■■■■■ ■■■■■■■■)	"The patient speaks in Telugu: '■■■■■ 6 ■■■■■■■■ ■■■■■■■■ ■■■■■■■■ ■■■■■■'. Watch Sarvam AI translate this live into clinical English: 'Bilateral knee joint pain for 6 months'."
0:30 - 0:40	Step 4: Click Upload / Camera Scan Rx	"The patient uploads a prior doctor's prescription. Sarvam Vision 3B extracts Telmisartan and flags High Serum Uric Acid at 7.8 mg/dL."
0:40 - 0:48	Click Confirm & Generate Token	"Token #061 is generated in under 1 second. Now let's switch to the Doctor Portal."
0:48 - 0:58	Open Doctor Portal (/doctor), tap #061	"Instantly synced! The doctor sees a pure English SOAP note, the patient's original audio with a Listen button, lab warnings, and Ayurvedic metrics. 1-Click approves to ABDM!"

Slide 6 • [5:45 - 6:00] • Closing Statement

"Jury members, healthcare in India does not have a shortage of clinical talent; it has a shortage of time. By taking history elicitation and document digitization off the physician's desk and placing it directly into the patient's hands, MediKiosk restores the sacred doctor-patient relationship. We are Team BINARY BRAINS, and we are ready to pilot MediKiosk with the Ministry of Ayush. Thank you, and we welcome your questions!"

■ Part 3: Anticipated Jury Q&A; & Defense Cheat Sheet

Q1: How will voice recognition work in a noisy, chaotic hospital OPD?

We tackle noise through a 3-layer architecture: (1) Hardware: Directional noise-canceling mic arrays with WebRTC DSP. (2) Dual-Mode UI: Every voice question has large icon touch-chips for noisy moments. (3) Constrained Grammars: Dialogue manager uses structured clinical vocabularies rather than unconstrained open speech.

Q2: Why use Sarvam AI instead of OpenAI Whisper or GPT-4?

Two critical reasons: Sovereignty & Accuracy. DPDPA 2023 mandates Indian health data residency. Western models also have high error rates on code-mixed rural Indian dialects (e.g. Telugu mixed with medical English). Sarvam-105B and Bulbul v3 are indigenous sovereign models trained specifically for Indian linguistic nuances.

Q3: What if the AI hallucinates a wrong symptom or makes an incorrect diagnosis?

MediKiosk is engineered with a strict Doctor-in-the-Loop fail-safe. The AI NEVER diagnoses or prescribes autonomously. It acts purely as a clinical scribe drafting a SOAP summary. When the doctor opens the portal, every field is 100% editable before saving to the official EHR.

Q4: How does your system capture deep Ayurvedic Dashavidha Pariksha parameters?

In AYUSH mode, the engine runs an adaptive ontology derived from the Charaka Samhita. Lifestyle questions map mathematically to Prakriti, Agni (Manda/Tikshna), Koshtha, and Vikriti. These parameters are standardized using national NAMASTE and WHO ICD-11 Traditional Medicine Module 2 (TM-2) coding.

Q5: What happens if a patient arrives with a life-threatening emergency like a heart attack?

We have an active Red-Flag Engine. If the patient utters emergency keywords (crushing chest pain, left arm pain, breathlessness) or presses the prominent red Zero-Click Emergency button, the kiosk bypasses routine queuing, triggers a STAT Red Alert, sounds an alarm on the Emergency Triage Console, and alerts the crash cart team.

Q6: How do you handle patients who are completely illiterate and cannot read?

MediKiosk has zero literacy barriers. Every screen is fully audio-guided with automatic native TTS. The UI uses high-contrast iconography, color-coded visual body maps, and conversational voice guidance. An elderly village patient can complete intake purely by talking to the kiosk naturally.

Q7: How is patient data protected under DPDPA 2023 and ABDM standards?

We enforce a Consent-First architecture. Patients authenticate via ABHA and listen to an audio explanation before granting granular consent. Data in transit uses TLS 1.3 encryption, and data at rest in PostgreSQL is protected by Supabase Row-Level Security (RLS). Temporary kiosk session data is wiped on submission.

Q8: How does your OCR handle messy, illegible doctor handwriting on prescriptions?

We use Sarvam Vision 3B VLM. Unlike generic OCR, a Vision-Language Model understands clinical context, recognizing medical terminology, dosages, and standard abbreviations (OD, BD, SOS, Tab, Syp). If confidence is below 85%, the raw document is highlighted for physician review.

Q9: What is your business model and post-hackathon sustainability plan?

MediKiosk operates on a B2G and B2B hospital model: (1) Government/AYUSH procurement under the National Ayush Mission & Ayush Grid with annual maintenance. (2) Private Hospital SaaS per-OPD-encounter tier. (3) Software licensing for existing state e-Hospital kiosk hardware without buying new machines.

Q10: Can your system function if the hospital internet goes down?

Yes. MediKiosk features an Offline/Async Resilient Fallback. The kiosk maintains an encrypted SQLite local cache on the device. Patients can continue voice intake, token generation, and local queue assignment offline. When connectivity is restored, data auto-syncs with the cloud database.